

Estimation (Part 2)

STAT 4530/6530
Statistical Inference for Data Scientists

Maximum Likelihood Estimation

Let $\mathbf{y} = (y_1, \dots, y_n)$ denote a vector of n observations, and let $f(\mathbf{y}; \theta)$ denote their joint probability function, according to a parametric family of probability distributions with parameter θ .

- Note: θ can be multidimensional, which is equivalent to the parameter θ actually being made up of multiple parameters.

Once we observe specific values for $\mathbf{y}$, we can plug those values into the joint probability function $f(\mathbf{y}; \theta)$. After doing so, $f(\mathbf{y}; \theta)$ depends only on the value of the parameter θ , so it can be viewed as a function of θ , and we refer to as the **likelihood function** and denote by $L(\theta)$.

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We typically assume that the observations $y_1, \dots, y_n$ are independent, and each observation y_i is distributed according to the same probability function $f(y, \theta)$. In this case,

$$L(\theta) = f(\mathbf{y}; \theta) = f(y_1; \theta) * f(y_2; \theta) * \dots * f(y_n; \theta) = \prod_{i=1}^n f(y_i; \theta).$$

Likelihood Function: Example

Suppose we observe n independent binary trials, so each y_i follows a $\text{Ber}(p)$ distribution with probability mass function defined by $f(1; p) = p$ and $f(0; p) = 1 - p$.

The likelihood function $L(p)$ is:

$$L(p) = f(\mathbf{y}; p) = \prod_{i=1}^n f(y_i; p) = \prod_{i=1}^n p^{y_i} (1 - p)^{1 - y_i} = p^{\sum_{i=1}^n y_i} (1 - p)^{n - \sum_{i=1}^n y_i}.$$

Interestingly, to calculate $L(p)$, we do not need to know the individual values of the y_i , merely the sum $\sum_{i=1}^n y_i$ (total number of “successes” in n trials).

Maximum Likelihood: Example

Recall that the goal of maximum likelihood estimation is to find the parameter value for which the observed data are most likely to occur. This can be accomplished by finding the parameter value that maximizes the likelihood function.

Is this a reasonable approach? Consider a data set of $n = 10$ trials with a total of $\sum_{i=1}^{10} y_i = 0$ successes.

- How plausible is it to observe 0 successes in n trials if the probability of a success is $p = 0.8$ vs., say, $p = 0.2$?

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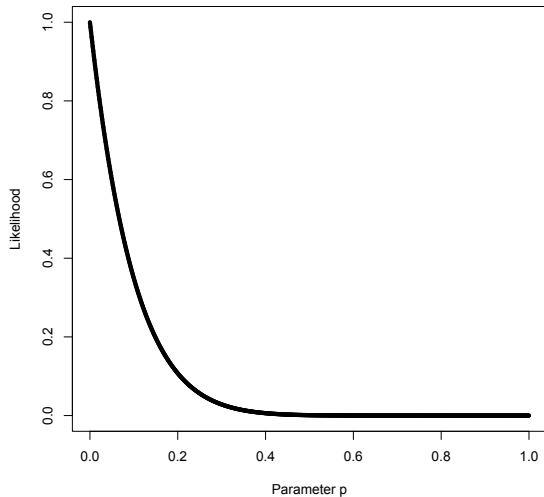
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- $L(0.8) = 0.0000001024$ and $L(0.2) = 0.1073742$
- If we make a plot of $L(p)$, we will see that it is maximized at $p = 0$. This makes sense: 0 successes in $n = 10$ trials is more likely to occur when the probability of a success is $p = 0$ than when p is equal to any other value.

Maximum Likelihood: Example



Maximum Likelihood: Example

It is not always easy to read off the maximum likelihood estimate by plotting the likelihood function. Instead, we can use calculus.

It is generally easier to work with the log-likelihood $l(\theta) = \log L(\theta)$:

- Because the log function is monotonically increasing, finding a value of θ that maximizes $L(\theta)$ is equivalent to finding a value of θ that maximizes $l(\theta)$
- The log-likelihood is easier to differentiate
- The log-likelihood is more numerically stable because it does not involve multiplying together many small values that could result in a number that is too small to represent on a computer

Exercise: For n binary trials, find the value of p that maximizes $l(p)$.

Properties of Maximum Likelihood Estimators

Maximum likelihood estimators are especially nice because they have desirable asymptotic (meaning, for a large sample size n) properties. In particular, they are:

- Asymptotically unbiased: as n increase, the bias gets closer to 0
- Consistent: as n increases, the estimator converges towards the parameter value
- Asymptotically efficient: for large n , other estimators do not tend to fall closer to the parameter and have smaller standard errors

Maximum likelihood estimators also have asymptotically normal sampling distributions.

Fisher Information

For n independent and identically distributed random variables $Y_1, \dots, Y_n$, with corresponding probability function $f(y; \theta)$ (where θ is a univariate parameter), the **Fisher information**, denoted by $I(\theta)$, is:

$$I(\theta) = n * E \left[\left(\frac{\partial \log f(Y_1; \theta)}{\partial \theta} \right)^2 \right].$$

The expectation is taken with respect to Y_1 for fixed θ . Instead of Y_1 , we could equivalently use any other Y_i .

The Fisher information increases as the sample size n increase, and it can be shown that the approximate distribution of the maximum likelihood estimator $\hat{\theta}$ is $N(\theta, 1/I(\theta))$. Thus the variance of the maximum likelihood estimator decreases as the sample size (and Fisher information) increases.

Fisher Information

Any estimator that has variance that simplifies to $1/I(\theta)$ for large n is asymptotically efficient.

- In practice, we do not know the value of θ , but we can estimate $\text{Var}(\hat{\theta})$ by $1/(I(\hat{\theta}))$.

Under certain conditions, an alternative expression for the Fisher information that can also use the log-likelihood function $l(\theta)$ is:

$$I(\theta) = -n * E \left[\frac{\partial^2 \log f(Y_1; \theta)}{\partial \theta^2} \right] = -n * E \left[\frac{\partial^2 l(\theta)}{\partial \theta^2} \right].$$

Since the second derivative of a function describes curvature, this alternative expression indicates that for a large sample $\text{Var}(\hat{\theta})$ is smaller when we expect the log-likelihood function to be more highly curved.

Examples

Suppose Y is a random variable with a $\text{Ber}(p)$ distribution.

- Compute the Fisher information $I(p)$
- Find the approximate distribution of the maximum likelihood estimate $\hat{p}$ for a large sample size n

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Suppose $Y_1, \dots, Y_n$ are independent random variables with a $\text{Poisson}(\mu)$ distribution.

- Find the maximum likelihood estimator $\hat{\mu}$
- Find the approximate distribution of $\hat{\mu}$ for a large sample size n

Reference

Agresti, A., & Kateri, M. (2021). Foundations of Statistics for Data Scientists: With R and Python (1st ed.). Chapman and Hall/CRC.